

Trainings ▾

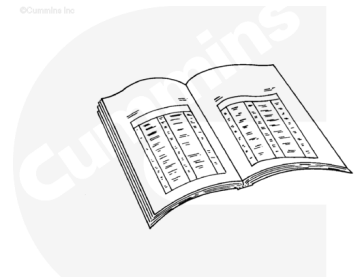
Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.



ck800wa

Note : The oil pump does not need to be removed.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>)
- Remove the oil pan adapter covers. Refer to Procedure 007-026 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-026-tr.html>)
- Remove the oil pan to gain access to the oil pump, if necessary. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html>)

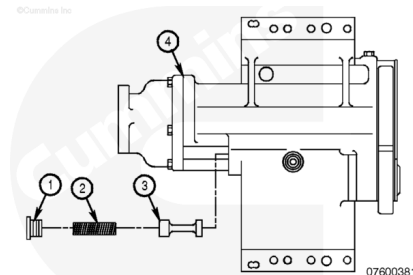
Remove

⚠ WARNING ⚠

The threaded plug (1) is under spring-pressure. To reduce the possibility of personal injury, wear proper eye protection when removing the threaded plug.

Note : The procedures for removing the oil pressure regulator from the oil pump on the QSK45 and QSK60 engines are the same. The oil pump for the QSK60 is shown.

Remove the threaded plug (1), spring (2), and pressure regulator plunger (3) from the oil pump (4).



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

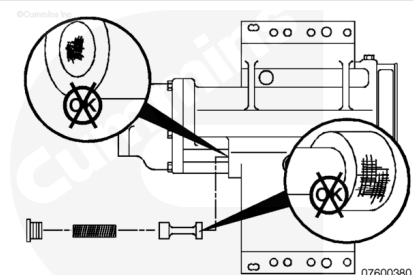
Note : The procedures for cleaning and inspecting the oil pressure regulator on the QSK45 and QSK60 engines are the same. The oil pressure regulator for the QSK60 is shown.

Clean the threaded plug, spring, and pressure regulator plunger with solvent.

Dry all of the parts with compressed air.

Inspect all of the oil pressure regulator parts for reuse.

Put the plunger in the bore, and move it to check for free movement. If the bore is damaged, the pump **must** be replaced. If the plunger is damaged, it **must** be replaced.

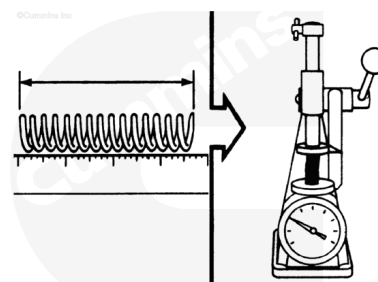


Check the free length of the spring. Two sizes of spring have been used. Identify which spring is used.

The free length of the spring will be approximately 203.2 mm [8 in] or 127.3 mm [5 in].

Used springs can be shorter by approximately 12.7 mm [0.5 in].

Use a valve spring tester, Part Number 3375182, or equivalent, to measure the spring force at the working height.



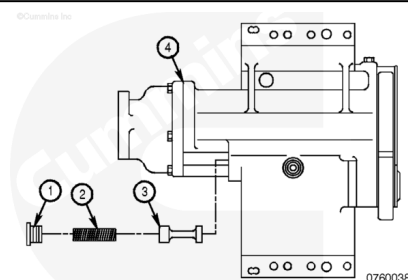
Working Height			
Spring Length		mm	in
127.3 mm [5 in]		70.1	2.76
203.2 mm [8 in]		113.8	4.48
Spring Force			
Spring Length	n		lbf
127.3 mm [5 in]	230	min	52
	255	max	57
203.2 mm [8 in]	249	min	56
	280	max	63

Install

Note : The procedures for installing the oil pressure regulator in the oil pump of the QSK45 and QSK60 engines are similar. The oil pump for the QSK60 is shown.

Use engine oil to lubricate the regulator plunger and bore.

Install the plunger (3) and spring (2) in oil pump (4) with threaded plug (1).



Torque Value: 136 n•m [100 ft-lb]

Finishing Steps

- Install the oil pan, if removed. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)

- Install the oil pan adapter covers. Refer to Procedure 007-026 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-026-tr.html)
- Fill the engine with clean 15W-40 oil to the correct level.
Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Operate the engine to check for oil pressure and leaks.

